Electronics 1 Course Project

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EE 3721

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Overall Description of the Project

This project involved the schematic design and implementation of a multi-output power supply system that provides several fixed and adjustable output voltage rails. The system includes input protection, rectification, regulation, and isolation features to ensure safe and stable operation.

The design process began with circuit simulation in Multisim, followed by implementing real world components in Altium then condensing into a PCB layout. After confirming the design, the circuit was built and tested on a breadboard, and then sent for PCB fabrication. Finally the board was soldered and tested.

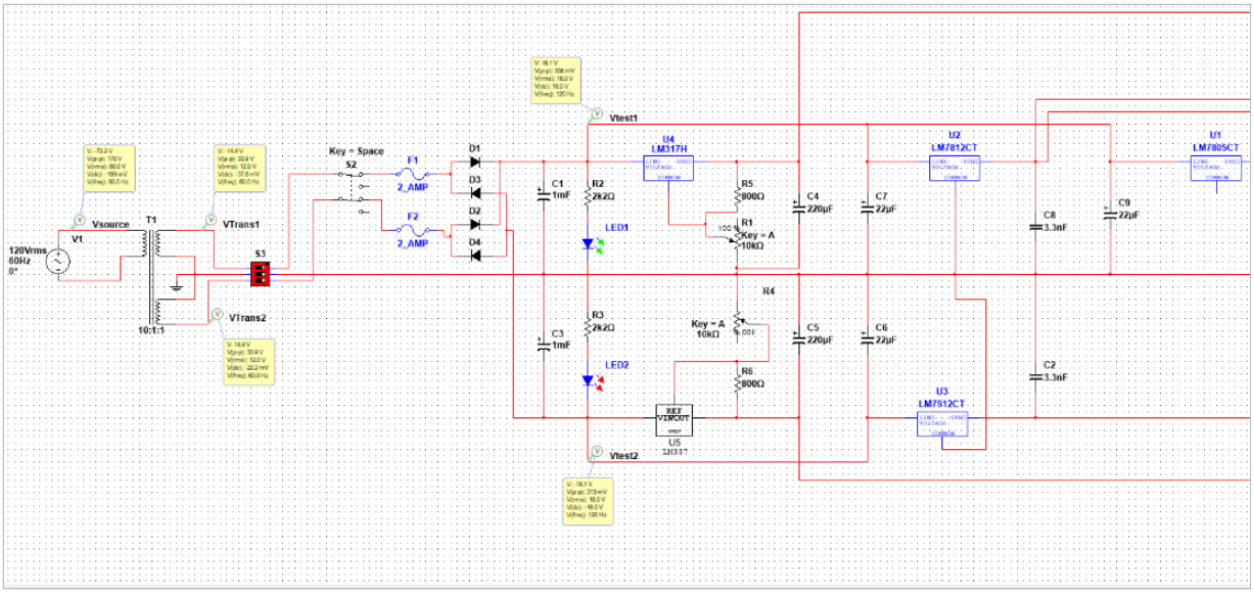


Figure 1: Schematic (first half)

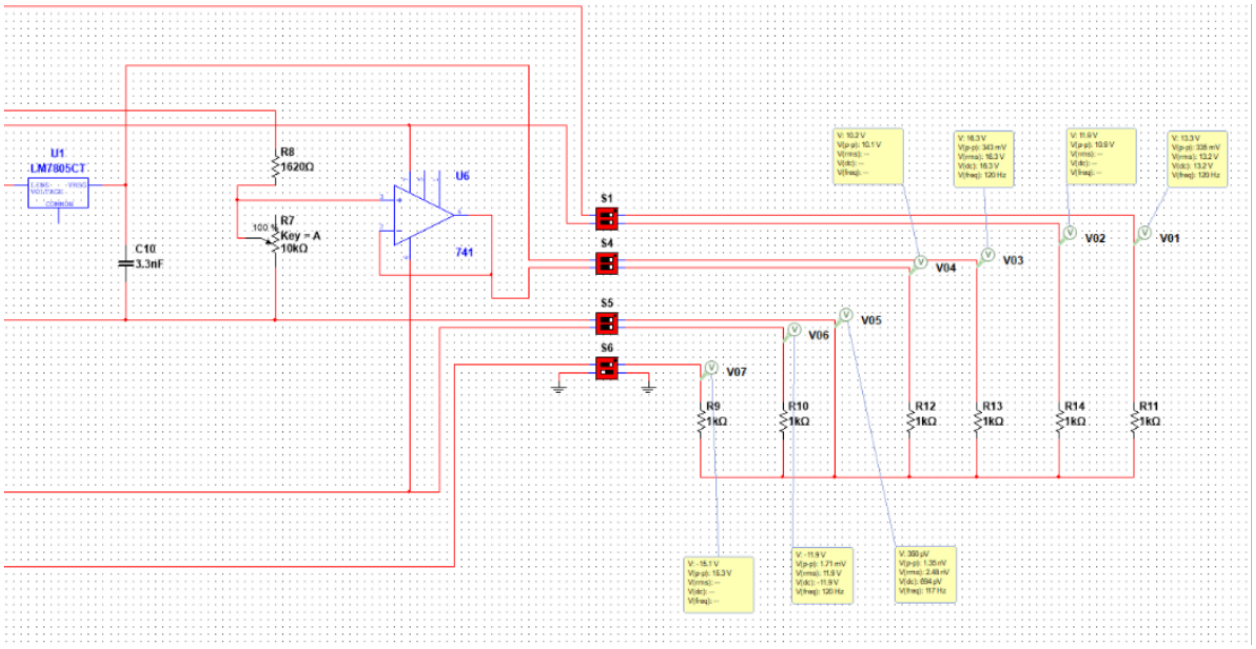


Figure 2: Schematic (second half)

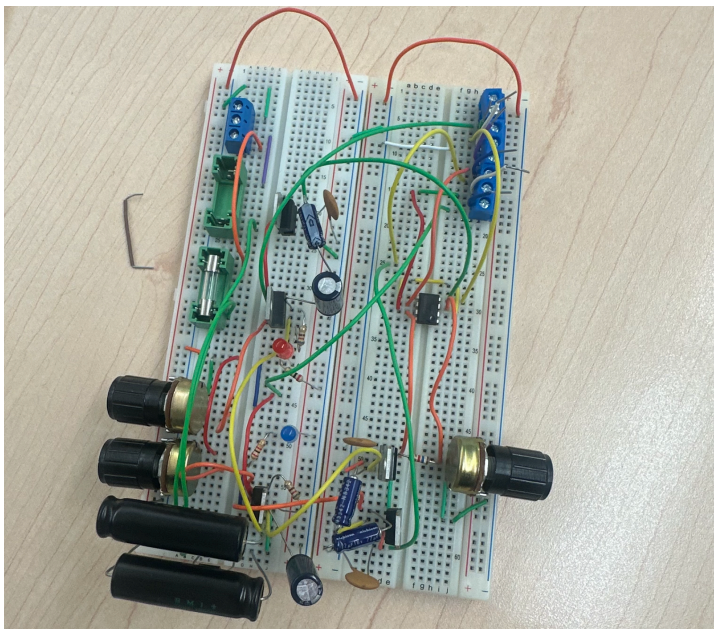


Figure 3: Circuit Prototype

Breakdown of Each Stage:

A. Inputs and Protection

The input stage accepts AC mains voltage and includes protective elements such as fuses and a switch. These components safeguard the circuit from overcurrent, voltage surges, and high-frequency noise, ensuring the rest of the system operates reliably. When the current is too high, short circuit, the fuse will blow which is important because it keeps the other, more expensive parts of our circuit safe from being damaged by too much current. The switch allows us to safely control when our system is on and off.

B. Rectification and Control

This stage converts the incoming AC voltage into DC using a full-wave bridge rectifier. After rectification, smoothing capacitors are used to reduce ripple. This section outputs two voltages for our output rail. A positive 1.5V to 19V and a -1.5V to 19V range. The range is controlled by B10K potentiometers as well as voltage regulators (LM317 for +1.5~19 and LM337 for -1.5~19) that allow for controlled but varying output voltages based on the resistivity of the potentiometer.

C. Power Supply (+12V)

This section provides a regulated 12V DC output to the output rail using the LM7812 voltage regulator. A 22uF capacitor is placed at the input to ensure smooth voltage into the voltage regulator, as well as a 3.3nF capacitor placed at the output to suppress high frequency noise. This allows a stable 12V for devices requiring 12V.

D. Power Supply (-12V)

This section is exactly the same as the 12V output but provides a regulated -12V DC output to the output rail. It uses a LM7912 voltage regulator. A 22uF capacitor is placed at the input to ensure smooth voltage into the voltage regulator, as well as a 3.3nF capacitor placed at the output to suppress high frequency noise. This allows a stable -12V for devices requiring -12V.

E. Power Supply (+5V)

This stage provides a +5V DC output using a UA7805 voltage regulator. It takes its input from the end of the rectifier, same as the +12V supply. It also uses a 22uF capacitor placed at the input to ensure smooth voltage into the voltage regulator, as well as a 3.3nF capacitor placed at the output to suppress high frequency noise. Ultimately providing a steady +5V to the output rail.

F. Adjustable Isolated Power Supply (+0~10V)

An isolated adjustable power supply is included to deliver a variable output from 0 to +10V. This is done with a UA741 OP amp used as a buffer. The B10K potentiometer is connected to the non-inverting terminal of the OP amp, allowing the output voltage to be adjustable. Pin 4 and 7 are connected to the output of the voltage regulators (LM7812 and LM7912) powering the OP amp. The buffer smoothes out the output voltage allowing an isolated adjustable 0 to +10V.

Results:

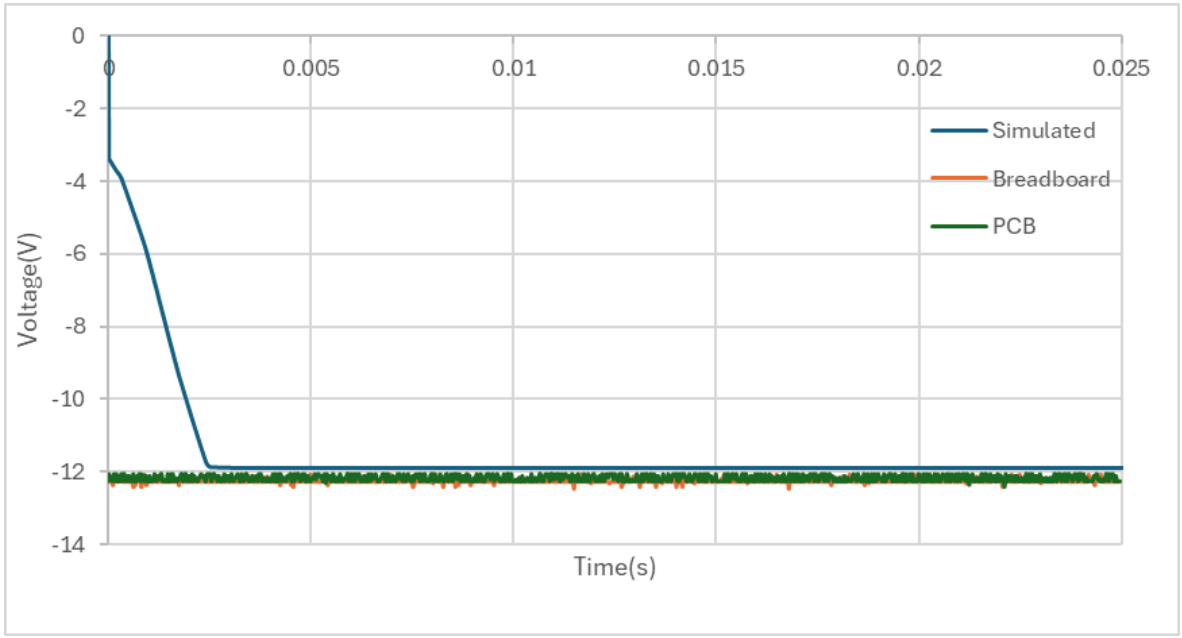


Figure 1: Output at -12V

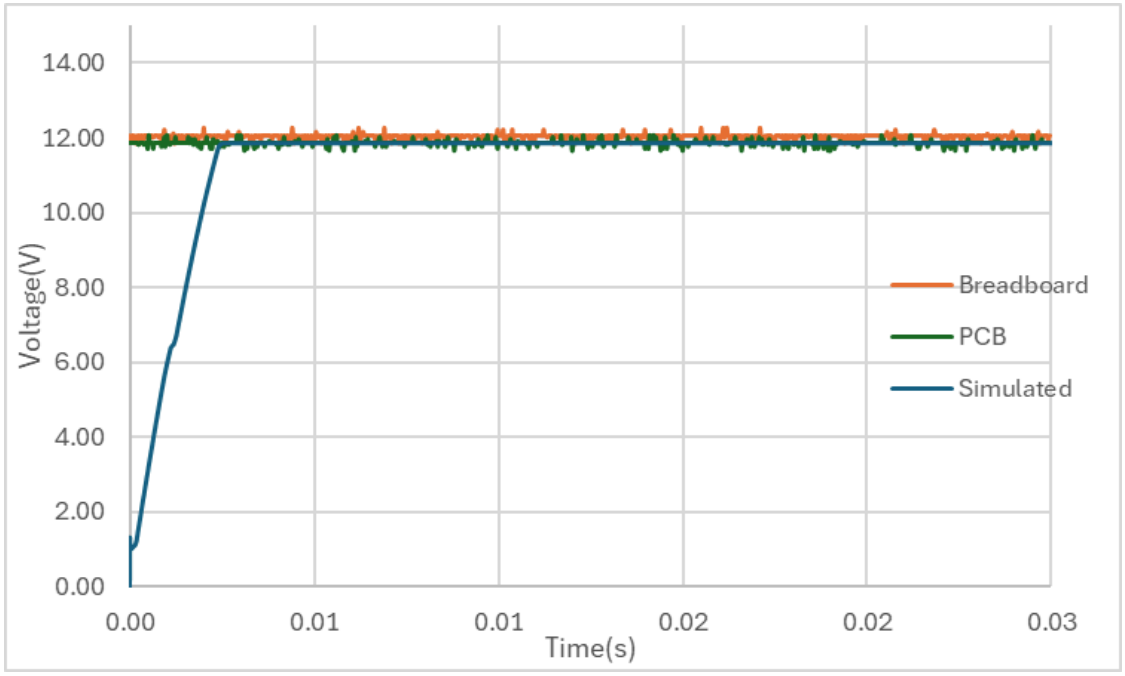


Figure 2: +12V

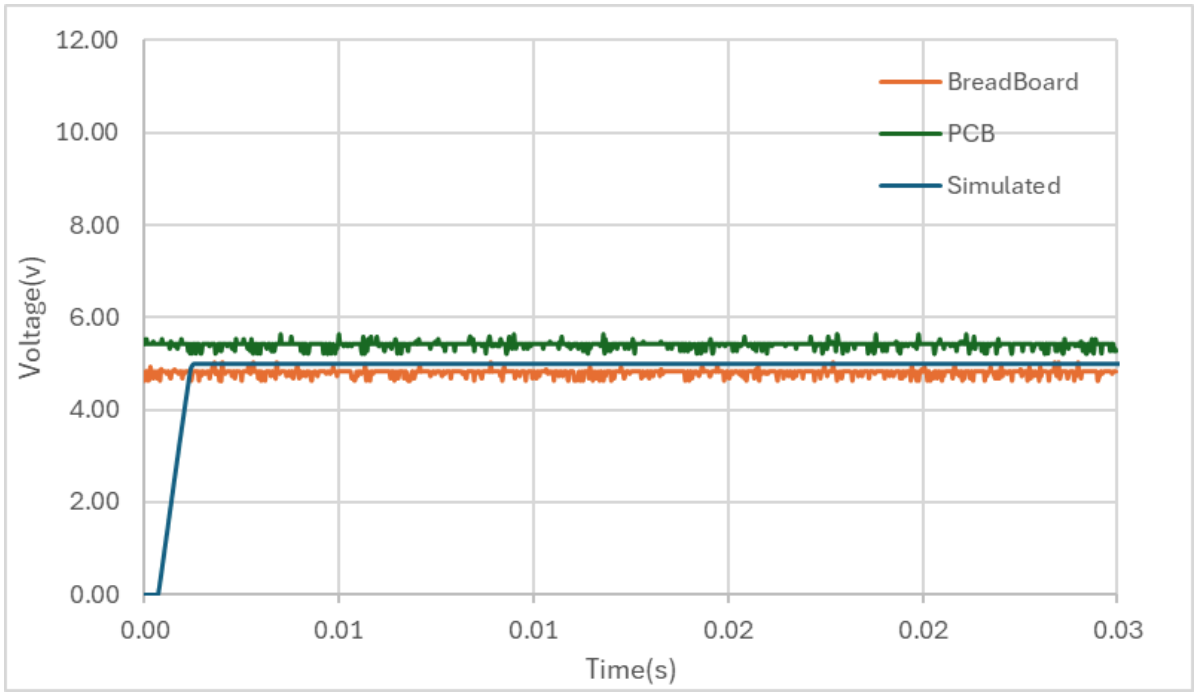


Figure 3: +5V

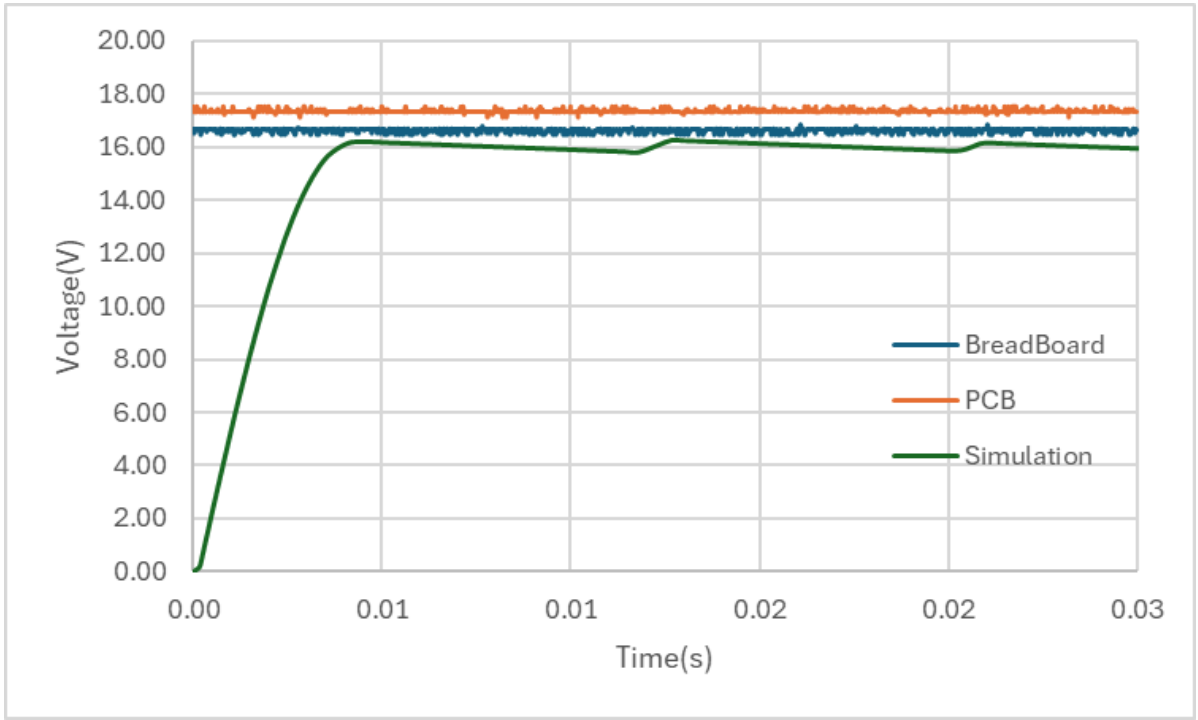


Figure 4: 1.5 to +19V (+19V)

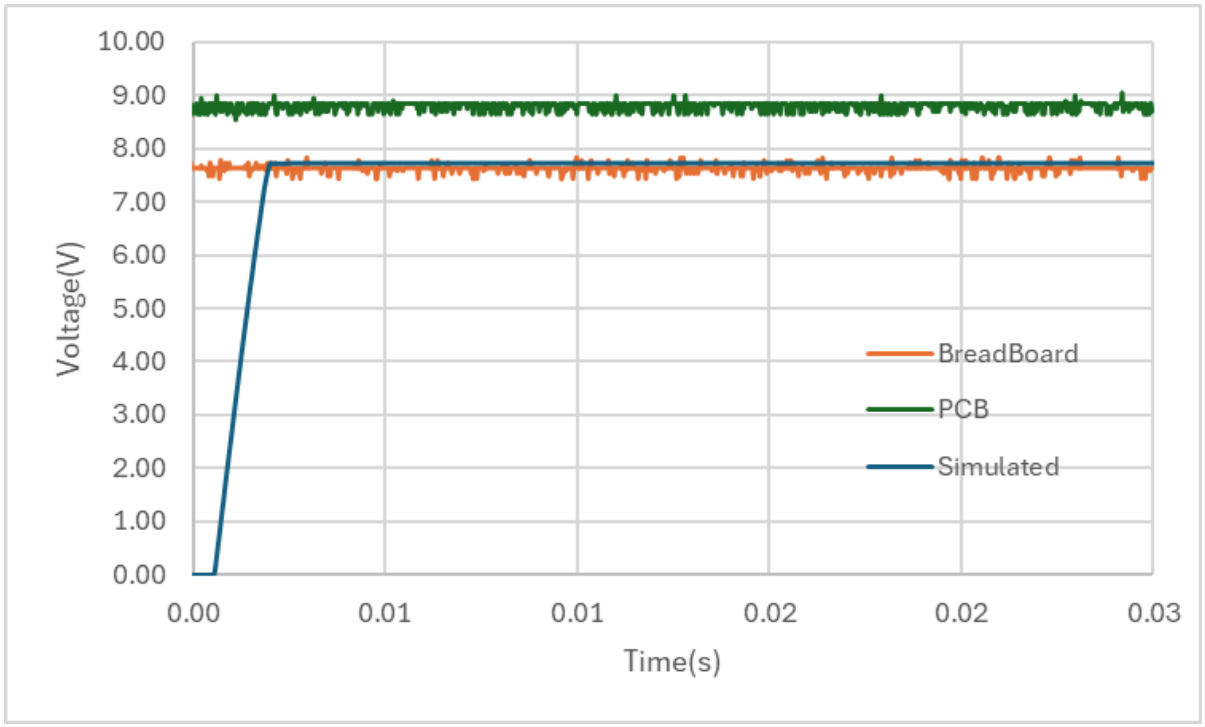


FIgure 5: 1.5 to 19V AVG (~8V)

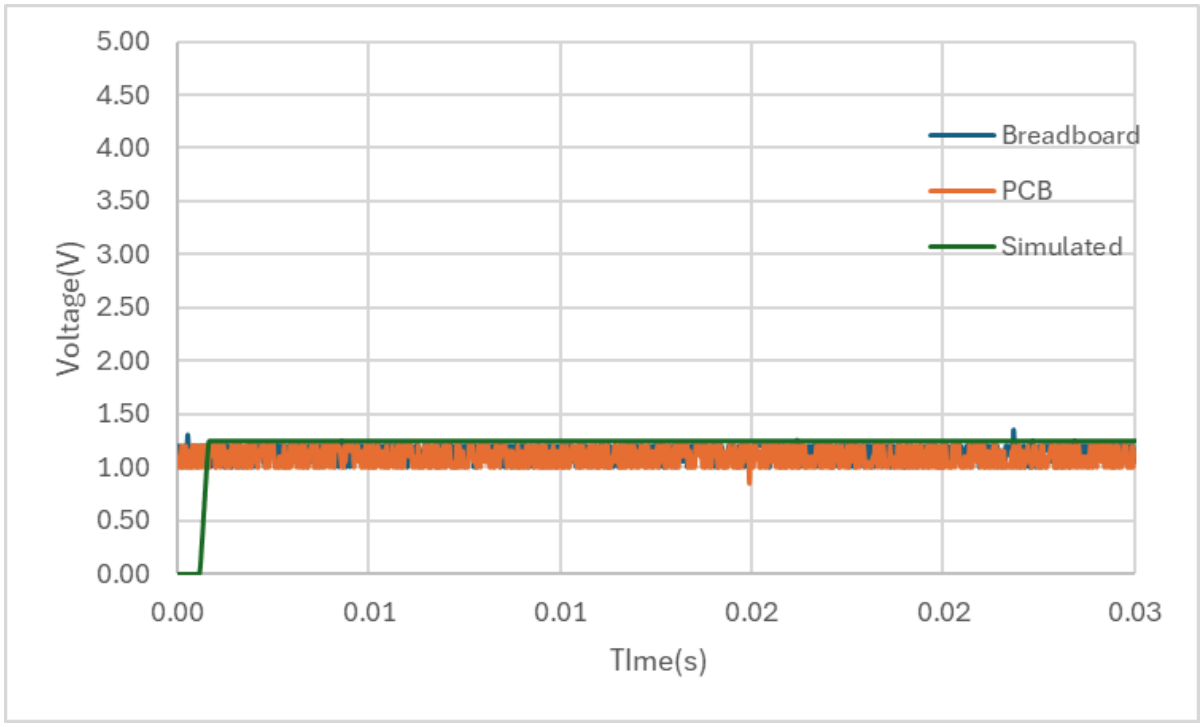


Figure 6: 1.5 to +19V (1.5V)

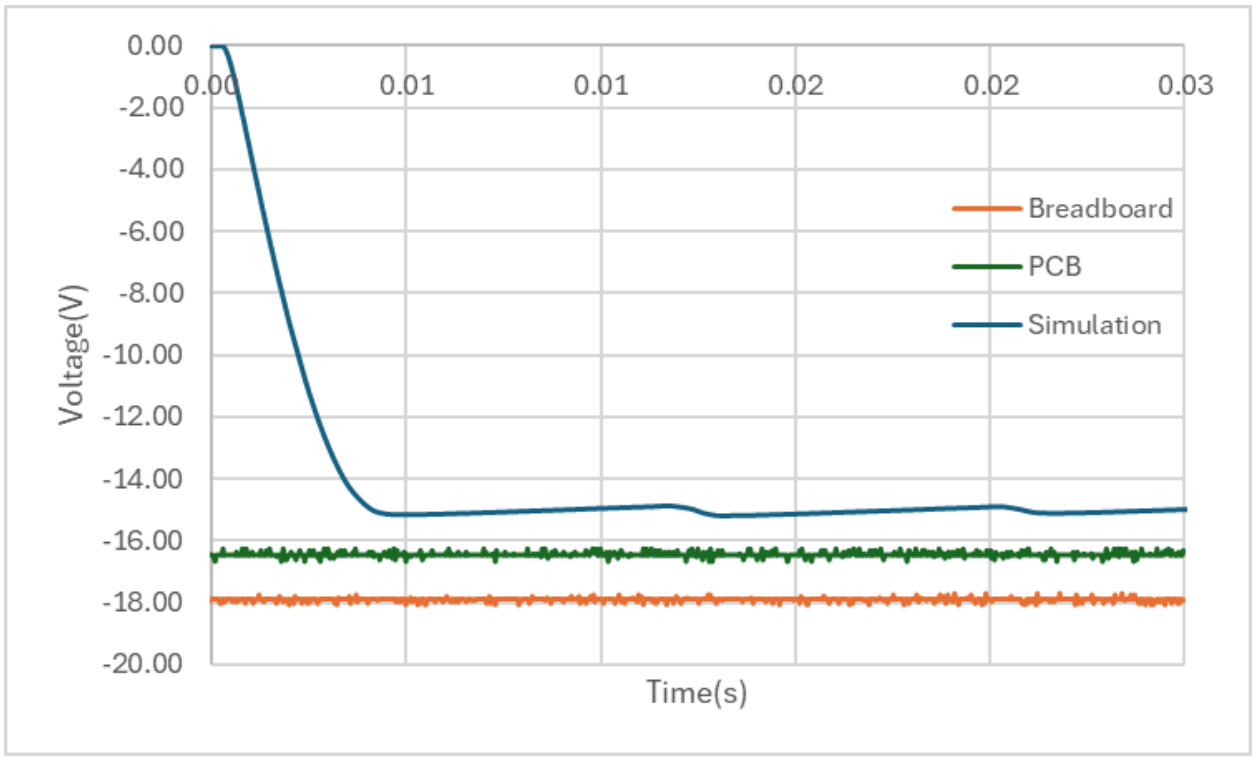


Figure 7: -1.5 to -19V (-19V)

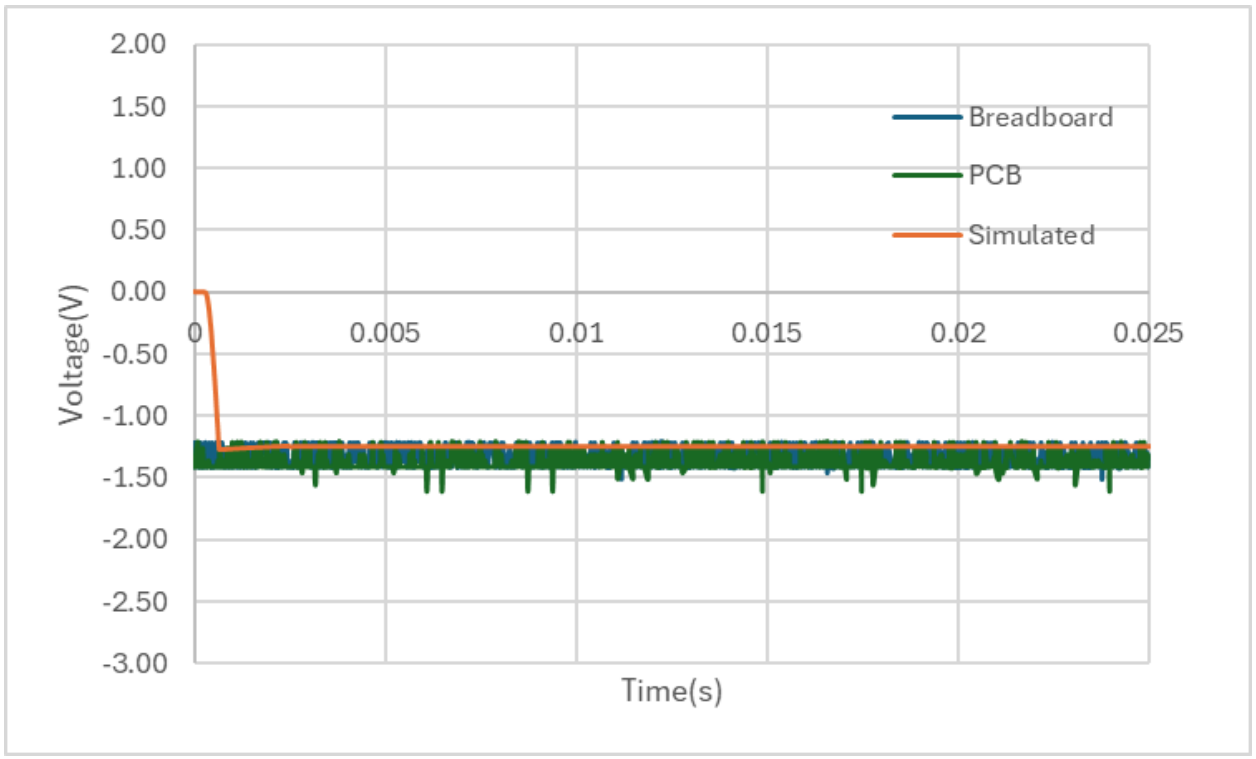


Figure 8: -1.5 to -19V (-1.5V)

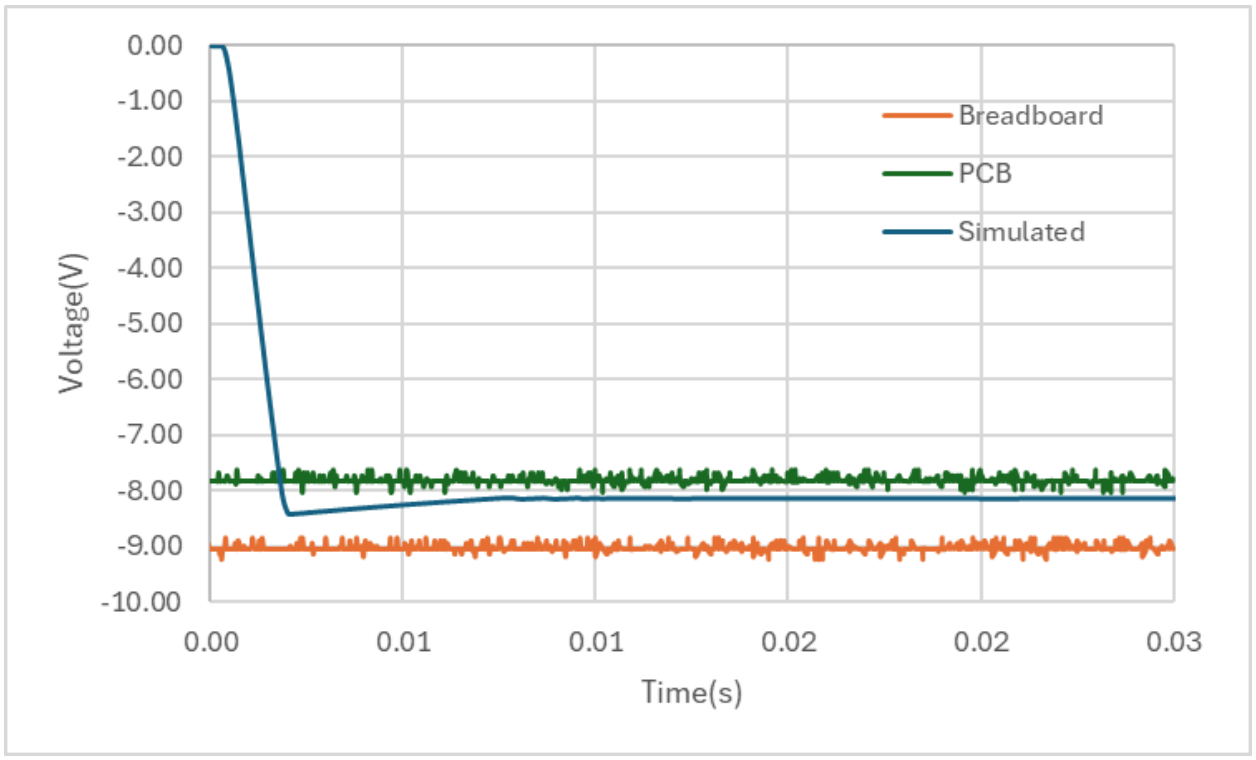


Figure 9: -1.5 to -19V AVG (~-8V)

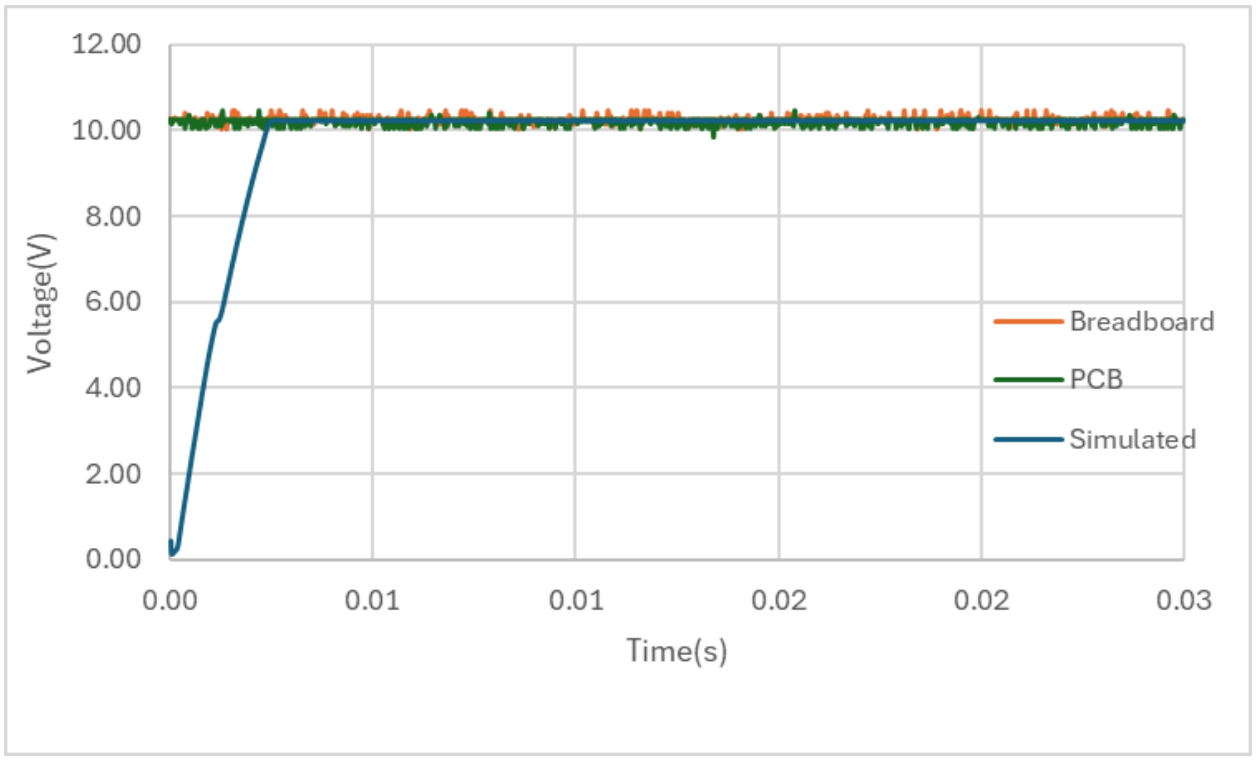


Figure 10: 0 to +10V (+10V)

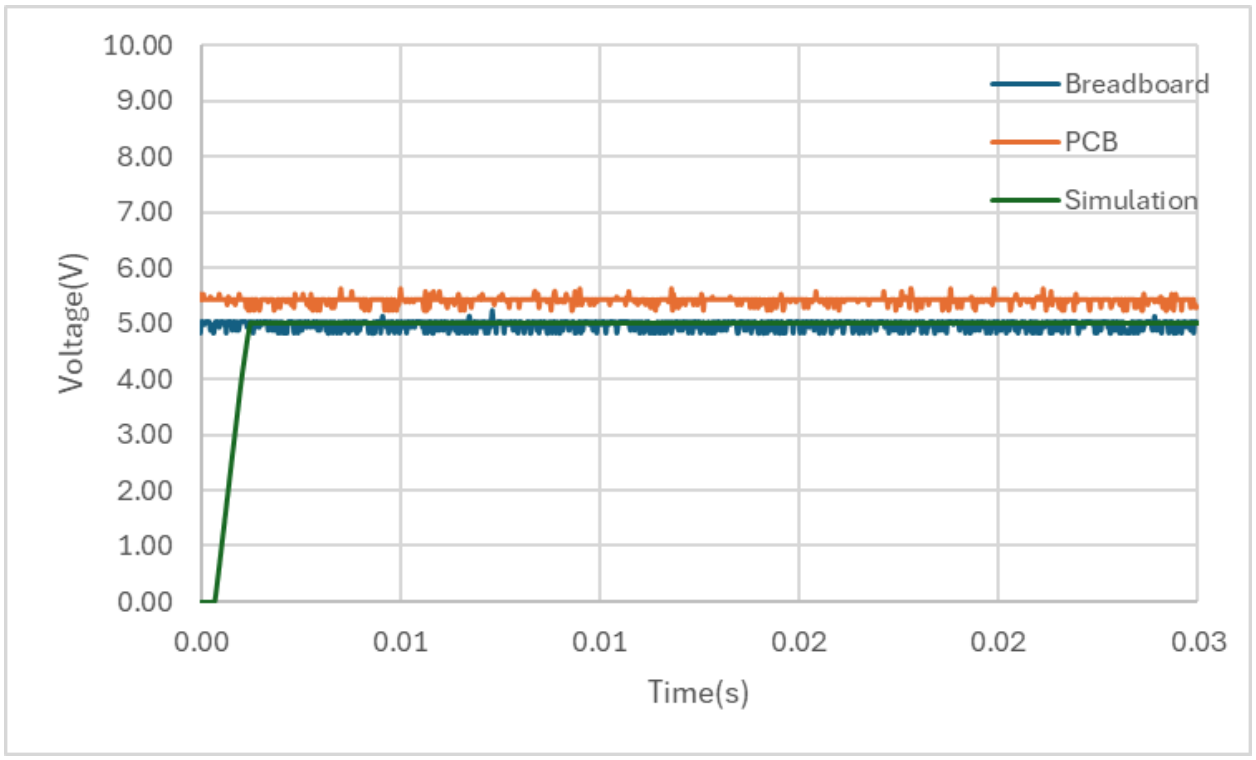


Figure 11: 0 to +10V AVG (~5V)

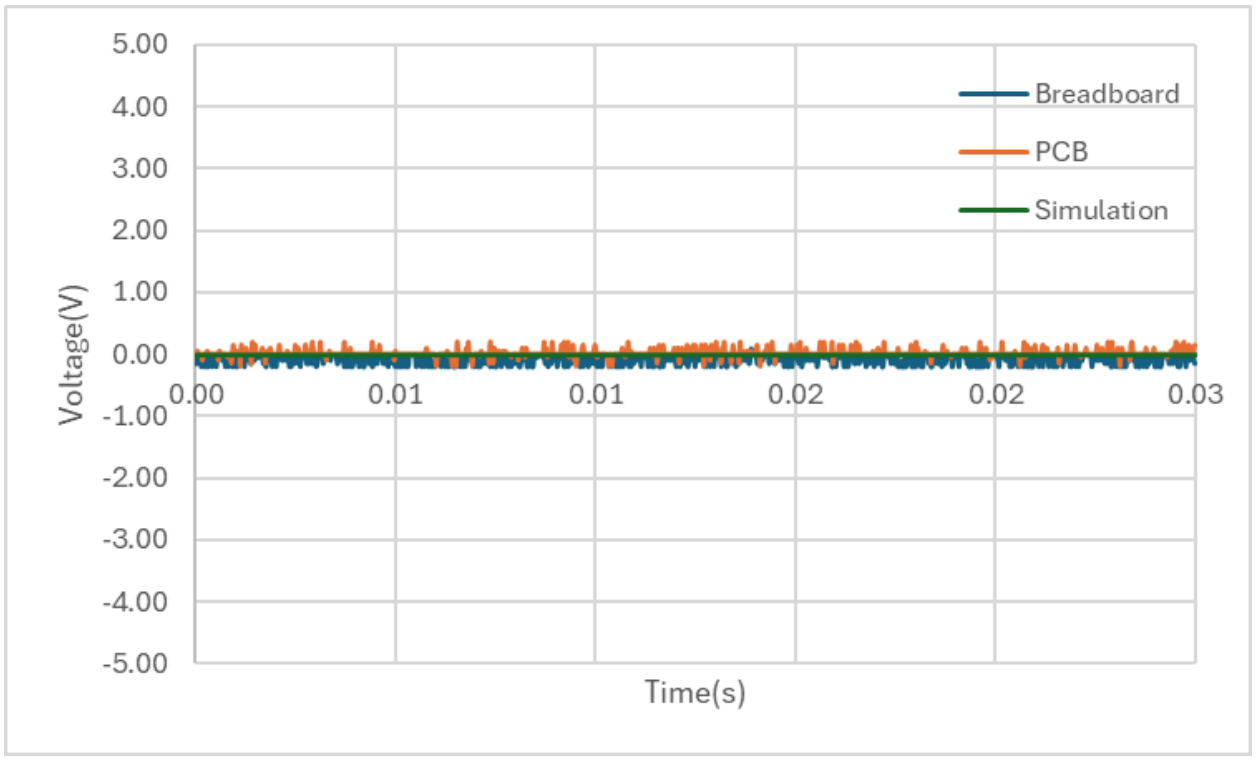


Figure 12: 0 to +10 V (0V)

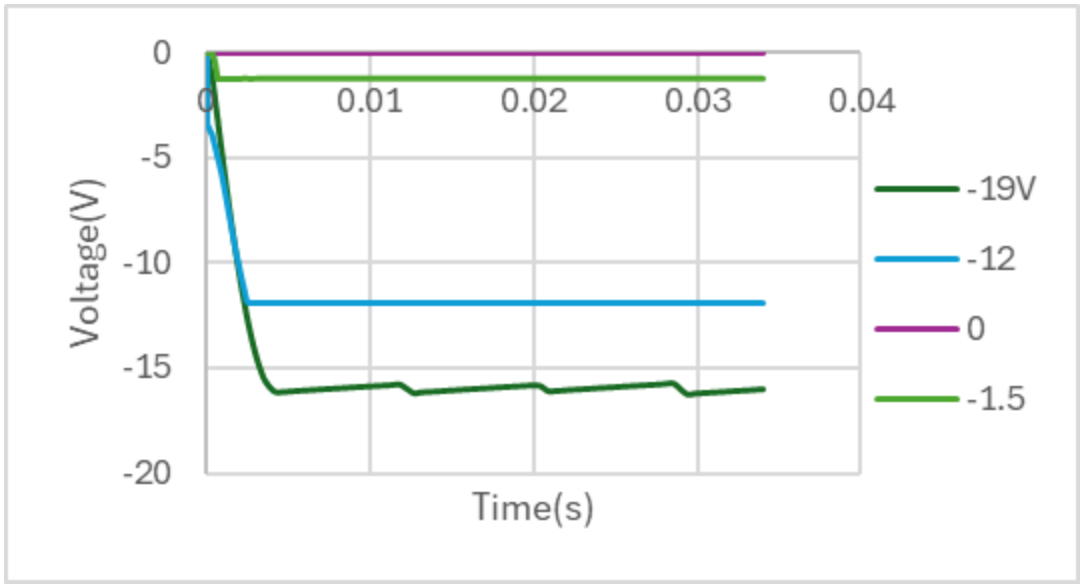


Figure 13: Negative Multisim outputs

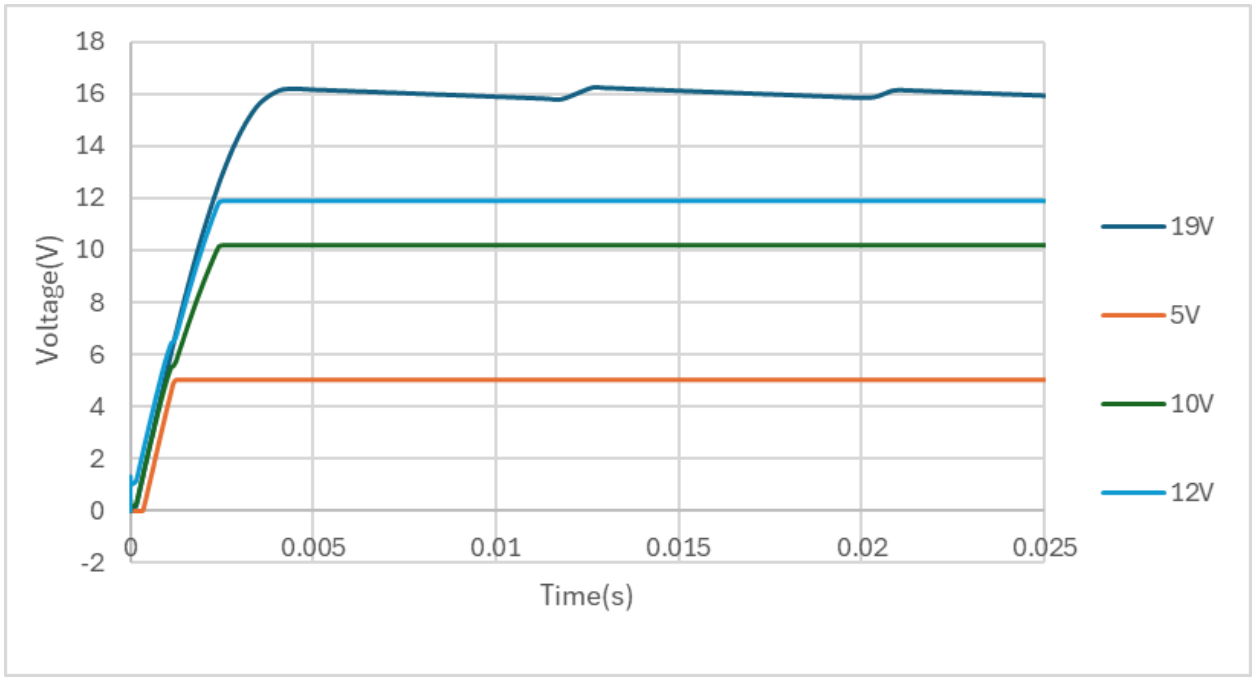


Figure 14: Positive Multisim outputs

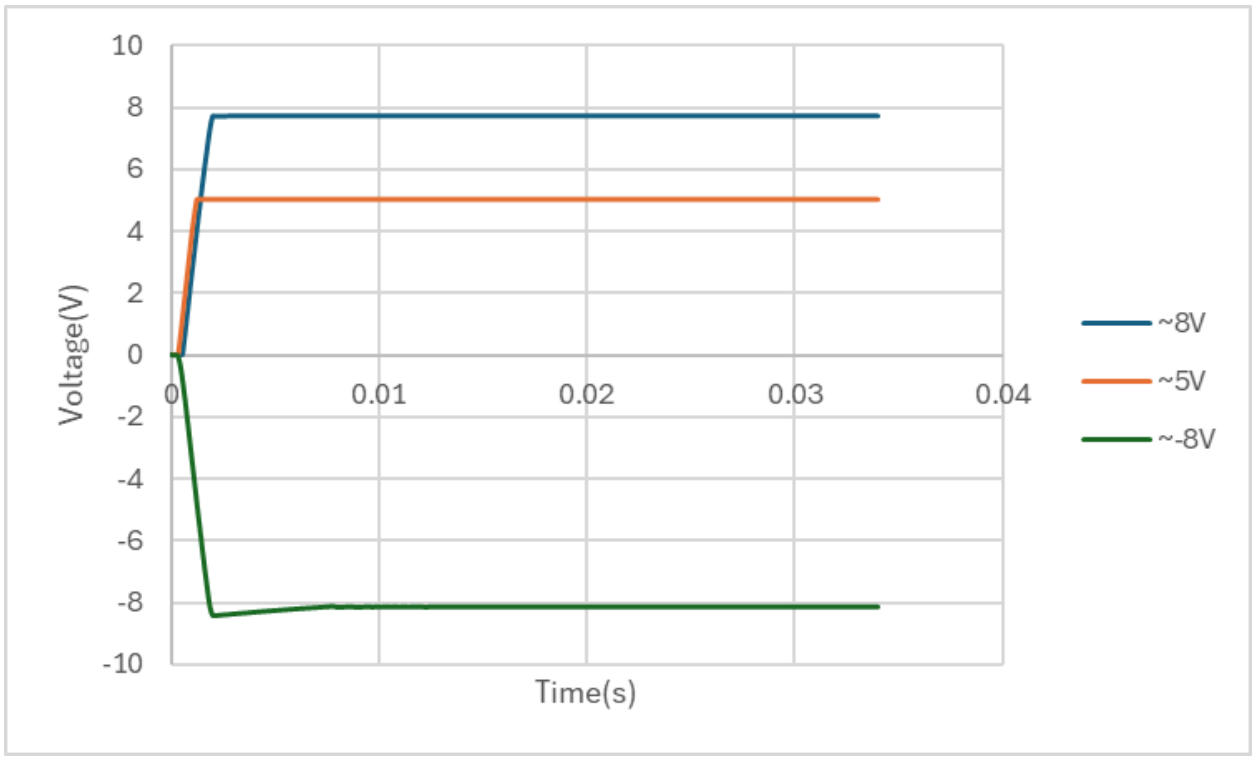


Figure 15: Average Multisim Data

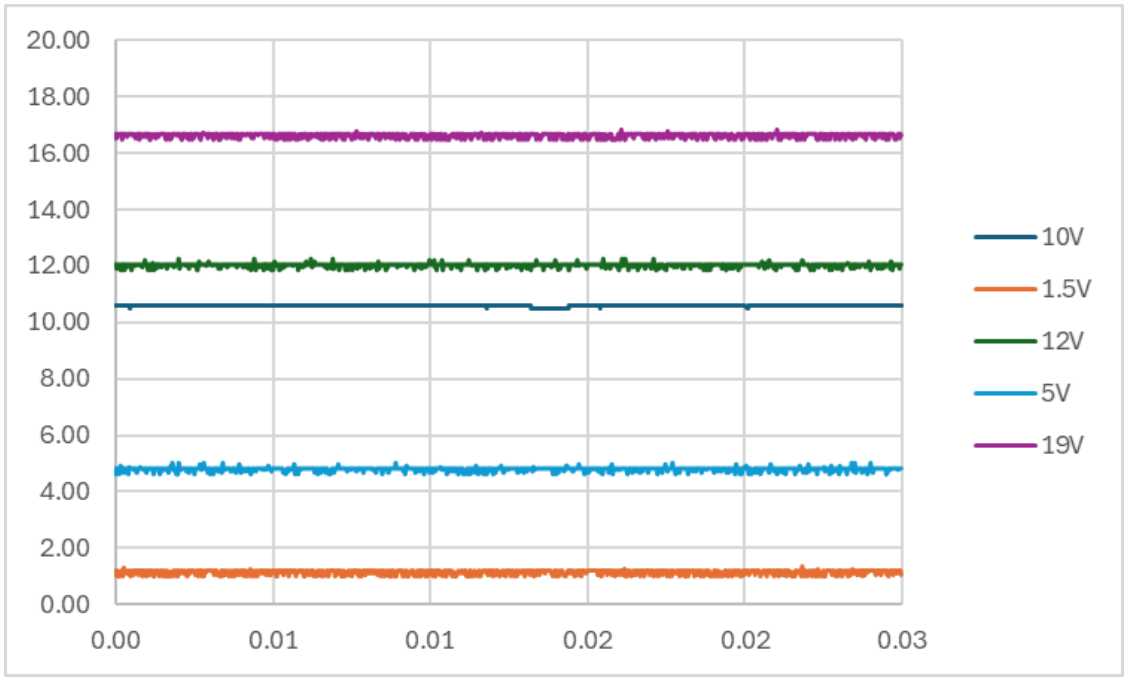


Figure 16: Positive Breadboard Data

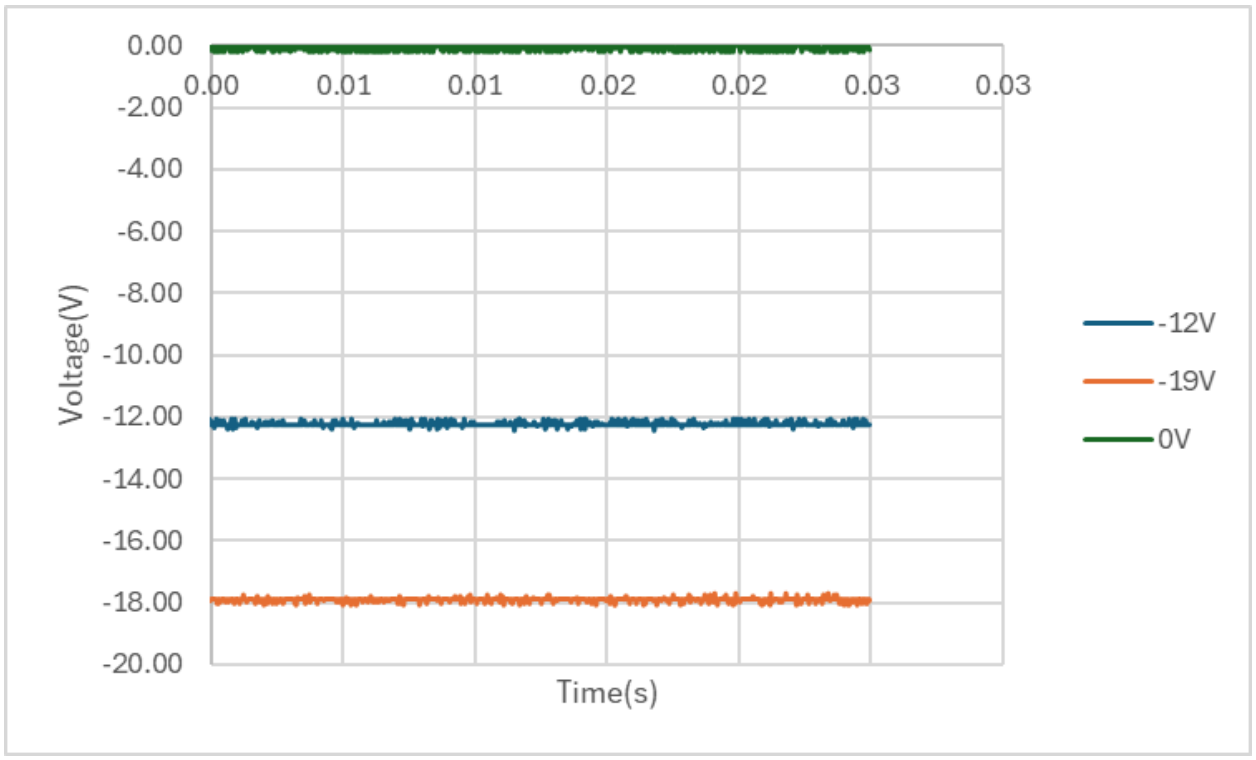


Figure 17: Negative Breadboard output

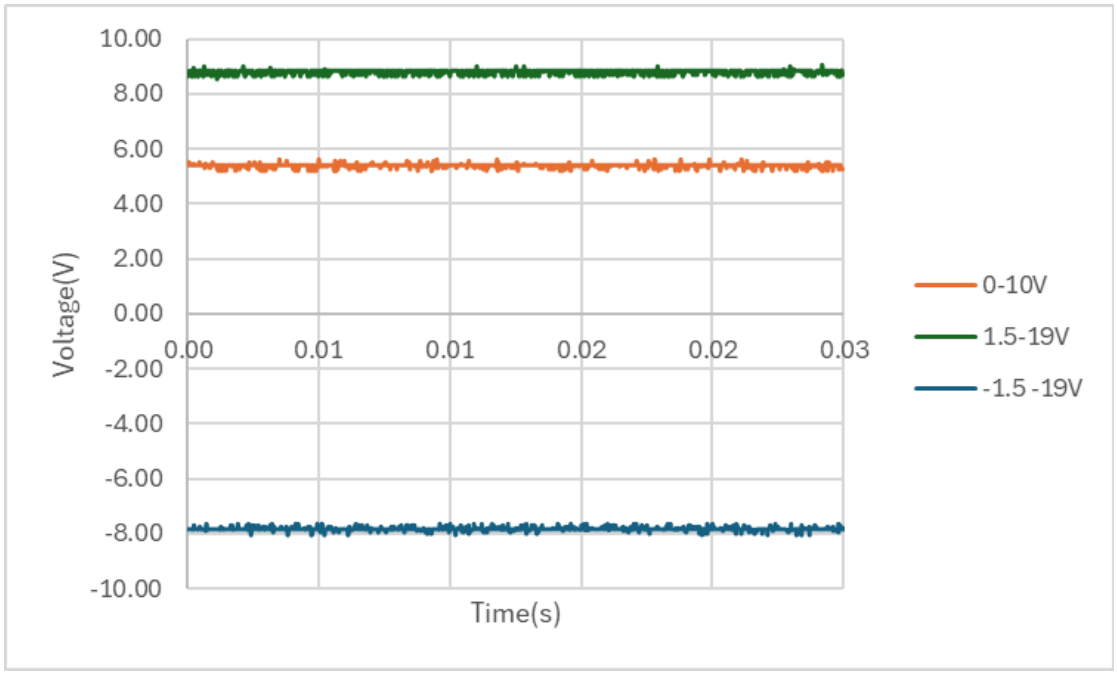


Figure 18: Average Breadboard outputs

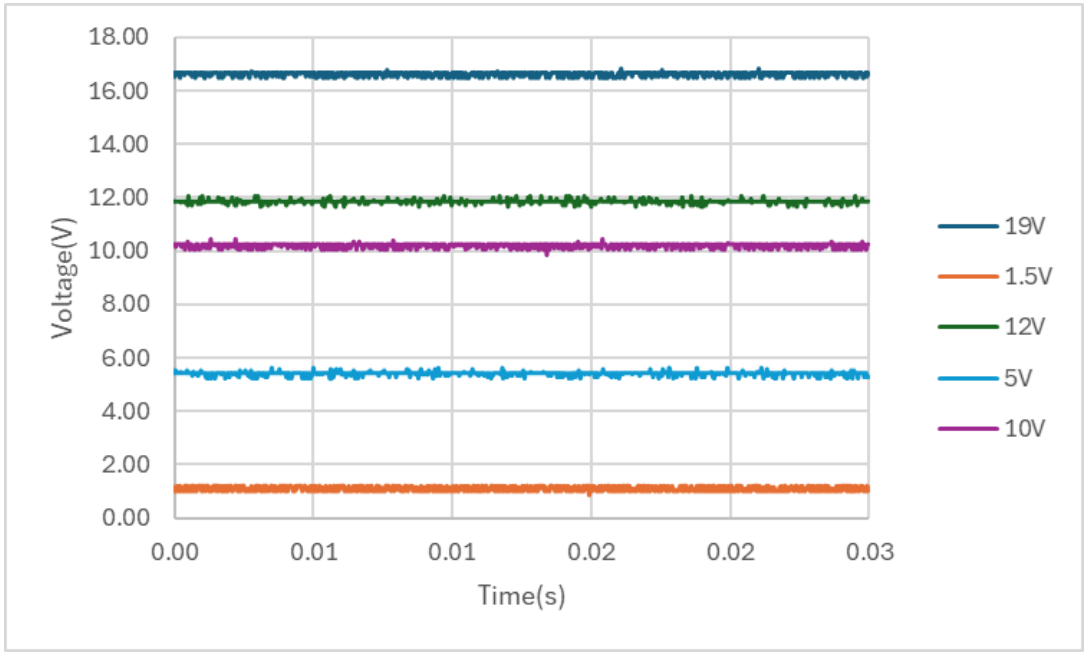


Figure 19: Positive PCB outputs

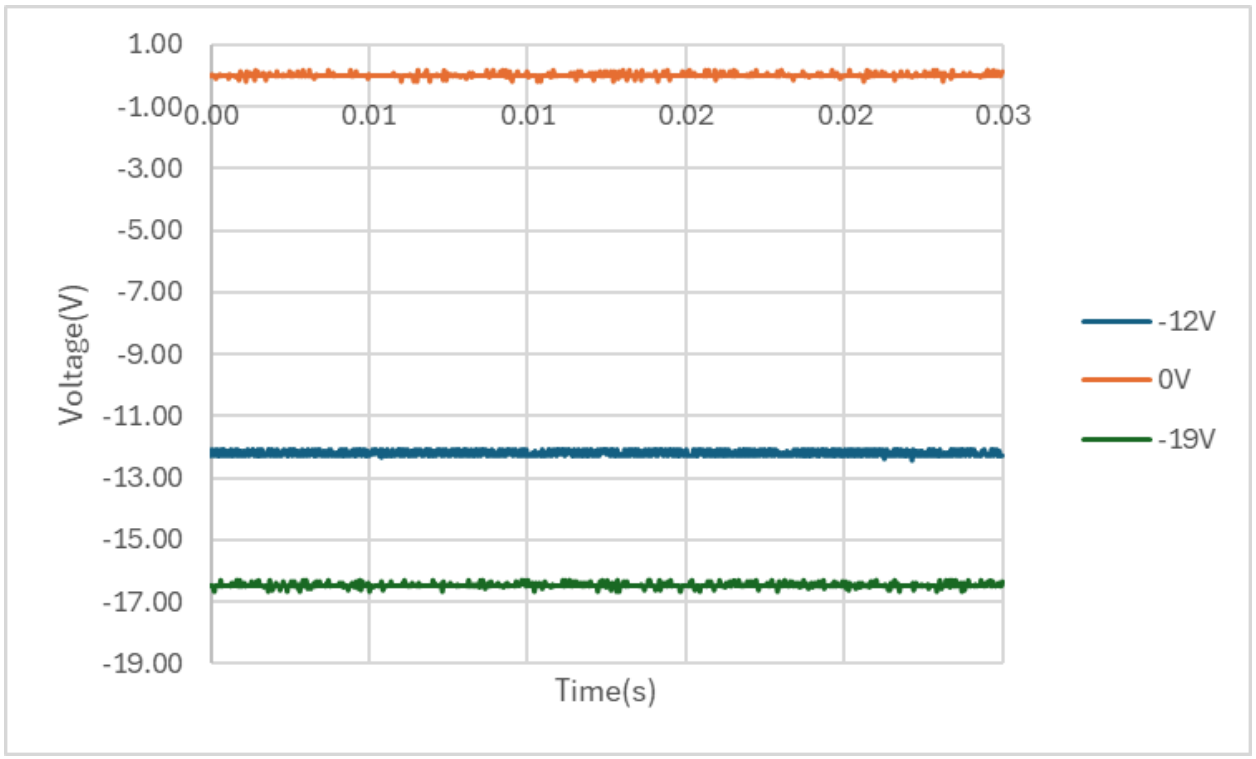


Figure 20: Negative PCB outputs

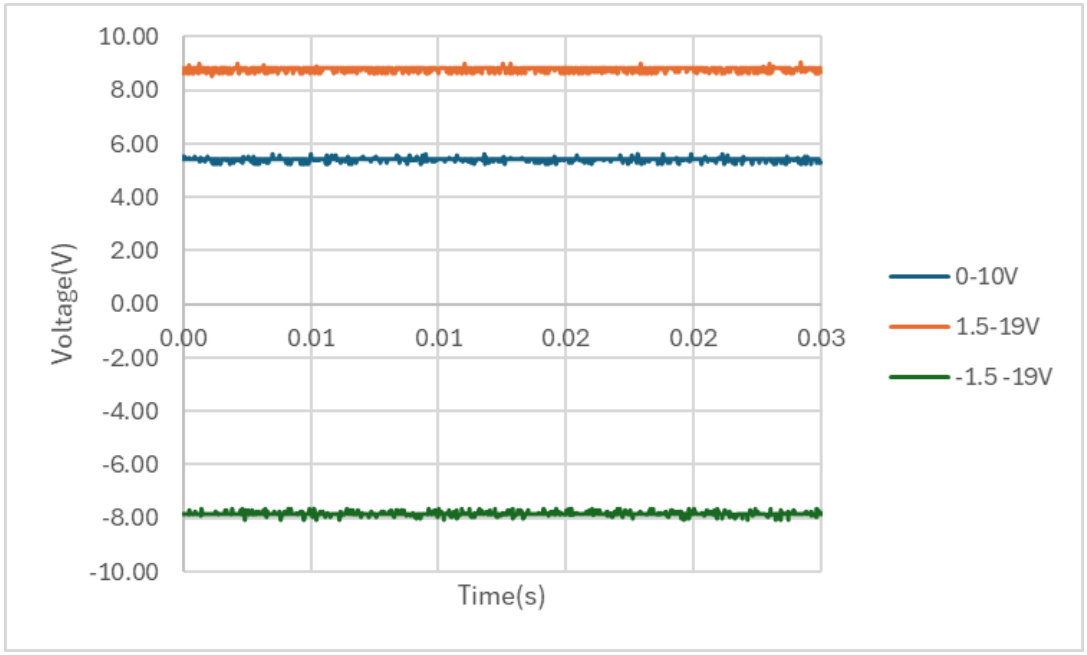


Figure 21: PCB average outputs

Analysis:

The output voltages for all regulated rails for Multisim simulation, breadboard, and PCB were overall consistent between the simulation, breadboard, and PCB tests indicating that the circuit was accurately built and working as expected. Each rail's output remained within expected ranges, with little noise in the PCB and breadboard data which is to be expected with physical materials and measurement vs simulation.

For fixed output rails such as +12V, –12V, and +5V, the values measured on the breadboard and PCB were nearly identical. Minor deviations were within acceptable voltage regulation tolerances and likely due to non-idealities.

For adjustable output rails (such as +0–10V, +1.5–19V, and –1.5–19V), the outputs were measured at minimum, maximum and average values. The breadboard and PCB data was as

expected and matched closely with the simulation data, validating the potentiometer control and op-amp buffering.

The only outliers were in the Multisim simulation data for the +19V and –19V rails. This was caused by Multisim not having the proper voltage regulators to get us closer to +19V and -19V. As a result, the simulated output did not reflect real-world performance and should be treated as an approximation rather than a definitive reference for these specific rails. Overall, the data was very consistent and proved the PCB and breadboard to each work as expected.

Appendix:

- Schematic of simulation circuit: Swartout_O_Schematic.pdf
- PCB design : Swartout_O_PCB.PcbDoc (has clearance errors but is accurate all traces 1mm)

- Bill of materials : Swartout_O_BillofMaterials.xlsx